

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 4 – SOLUTIONS

1. Suppose that the process $\{X_t\}$ is defined

$$X_t = a + bt + s_t + Y_t$$

where a and b are constants, s_t is a seasonal factor such that $s_t = s_{t-12}$ and $\{Y_t\}$ is a ‘noise’ process (note, $\{Y_t\}$ is not necessarily a white noise process).

We define $\{W_t\}$ to be the process:

$$W_t = (1 - B)(1 - B^{12})X_t.$$

Determine the ACF of $\{W_t\}$ for the following cases:

- (a) Where $\{Y_t\}$ is a white noise process with variance σ_z^2 .

SOLUTION:

We note that

$$\begin{aligned} W_t &= (1 - B)(1 - B^{12})X_t \\ &= (1 - B)(X_t - X_{t-12}) \\ &= (1 - B)(a + bt + s_t + Y_t - a - b(t - 12) - s_{t-12} - Y_{t-12}) \\ &= (1 - B)(12b + Y_t - Y_{t-12}) \\ &= Y_t + Y_{t-13} - Y_{t-1} - Y_{t-12}. \end{aligned}$$

If Y_t is a white noise process with variance σ_z^2 , then $Y_t = Z_t$ and

$$\begin{aligned} \text{Var}(W_t) &= \gamma(0) = \text{Var}(Z_t + Z_{t-13} - Z_{t-1} - Z_{t-12}) \\ &= \text{Var}(Z_t) + \text{Var}(Z_{t-13}) + \text{Var}(Z_{t-1}) + \text{Var}(Z_{t-12}) \\ &= 4\sigma_z^2 \end{aligned}$$

Now consider $\gamma(h) = \text{Cov}(W_t, W_{t+h})$ for $|h| > 0$.

$$\begin{aligned} \gamma(h) &= \mathbb{E}[(Z_t + Z_{t-13} - Z_{t-1} - Z_{t-12})(Z_{t+h} + Z_{t+h-13} - Z_{t+h-1} - Z_{t+h-12})] \\ &= \mathbb{E}(Z_t Z_{t+h}) + \mathbb{E}(Z_t Z_{t+h-13}) - \mathbb{E}(Z_t Z_{t+h-1}) - \mathbb{E}(Z_t Z_{t+h-12}) \\ &\quad + \mathbb{E}(Z_{t-13} Z_{t+h}) + \mathbb{E}(Z_{t-13} Z_{t+h-13}) - \mathbb{E}(Z_{t-13} Z_{t+h-1}) - \mathbb{E}(Z_{t-13} Z_{t+h-12}) \\ &\quad - \mathbb{E}(Z_{t-1} Z_{t+h}) - \mathbb{E}(Z_{t-1} Z_{t+h-13}) + \mathbb{E}(Z_{t-1} Z_{t+h-1}) + \mathbb{E}(Z_{t-1} Z_{t+h-12}) \\ &\quad - \mathbb{E}(Z_{t-12} Z_{t+h}) - \mathbb{E}(Z_{t-12} Z_{t+h-13}) + \mathbb{E}(Z_{t-12} Z_{t+h-1}) + \mathbb{E}(Z_{t-12} Z_{t+h-12}) \end{aligned}$$

Looking at the above expression, we see that non-zero terms occur where $h = 1, 11, 12$ and 13. For all other $|h|$, $\gamma(h) = 0$. Then

$$\begin{aligned}\gamma(1) &= -\mathbb{E}(Z_t^2) - \mathbb{E}(Z_{t-12}^2) \\ &= -2\sigma_z^2.\end{aligned}$$

$$\begin{aligned}\gamma(11) &= \mathbb{E}(Z_{t-1}^2) \\ &= \sigma_z^2.\end{aligned}$$

$$\begin{aligned}\gamma(12) &= -\mathbb{E}(Z_t^2) - \mathbb{E}(Z_{t-1}^2) \\ &= -2\sigma_z^2.\end{aligned}$$

$$\begin{aligned}\gamma(13) &= \mathbb{E}(Z_t^2) \\ &= \sigma_z^2.\end{aligned}$$

Using the definition $\rho(h) = \gamma(h)/\gamma(0)$, we see that

$$\rho(h) = \begin{cases} 1 & h = 0; \\ -\frac{1}{2} & |h| = 1; \\ \frac{1}{4} & |h| = 11; \\ -\frac{1}{2} & |h| = 12; \\ \frac{1}{4} & |h| = 13; \\ 0 & \text{otherwise.} \end{cases}$$

Alternative solution: use the autocovariance generating function.

(b) Where $Y_t = (1 + \theta B)Z_t$, here $\{Z_t\}$ is a white noise process with variance σ_z^2 .

SOLUTION:

From (a), we know that

$$W_t = Y_t + Y_{t-13} - Y_{t-1} - Y_{t-12}.$$

Here

$$\begin{aligned}Y_t &= (1 + \theta B)Z_t \\ &= Z_t + \theta Z_{t-1}.\end{aligned}$$

Then

$$\begin{aligned}W_t &= Z_t + \theta Z_{t-1} + Z_{t-13} + \theta Z_{t-14} - Z_{t-1} - \theta Z_{t-2} - Z_{t-12} - \theta Z_{t-13} \\ &= Z_t + (\theta - 1)Z_{t-1} - \theta Z_{t-2} - Z_{t-12} + (1 - \theta)Z_{t-13} + \theta Z_{t-14}.\end{aligned}$$

Clearly, $\gamma(0) = \mathbb{E}(W_t^2)$:

$$\begin{aligned}
\gamma(0) &= \mathbb{E}(Z_t^2) + (\theta - 1)^2 \mathbb{E}(Z_{t-1}^2) + \theta^2 \mathbb{E}(Z_{t-2}^2) + \mathbb{E}(Z_{t-12}^2) + (1 - \theta)^2 \mathbb{E}(Z_{t-13}^2) + \theta^2 \mathbb{E}(Z_{t-14}^2) \\
&= \sigma_z^2 [1 + (\theta - 1)^2 + \theta^2 + 1 + (1 - \theta)^2 + \theta^2] \\
&= \sigma_z^2 [4 - 4\theta + 4\theta^2] \\
&= 4\sigma_z^2 (1 - \theta + \theta^2).
\end{aligned}$$

Consider $\gamma(h) = \text{Cov}(W_{t-h}W_t)$, then

$$\gamma(h) = \mathbb{E}(W_{t-h}(Z_t + (\theta - 1)Z_{t-1} - \theta Z_{t-2} - Z_{t-12} + (1 - \theta)Z_{t-13} + \theta Z_{t-14}))$$

Since W_{t-h} is a function of $Z_{t-h}, Z_{t-h-1}, Z_{t-h-2}, Z_{t-h-12}, Z_{t-h-13}$ and Z_{t-h-14} , for $|h| > 0$, the only non-zero expectations when evaluating $\gamma(h)$, will be those where $|h| = 1, 2, 10, 11, 12, 13$ or 14 .

$$\begin{aligned}
\gamma(1) &= (\theta - 1)\mathbb{E}(Z_{t-1}^2) - \theta(\theta - 1)\mathbb{E}(Z_{t-2}^2) - (1 - \theta)\mathbb{E}(Z_{t-13}^2) + \theta(1 - \theta)\mathbb{E}(Z_{t-14}^2) \\
&= \sigma_z^2 (2(\theta - 1) + 2\theta(1 - \theta)) \\
&= -2\sigma_z^2 (\theta - 1)^2.
\end{aligned}$$

$$\begin{aligned}
\gamma(2) &= -\theta\mathbb{E}(Z_{t-2}^2) - \theta\mathbb{E}(Z_{t-14}^2) \\
&= -2\theta\sigma_z^2.
\end{aligned}$$

$$\begin{aligned}
\gamma(10) &= \theta\mathbb{E}(Z_{t-12}^2) \\
&= \theta\sigma_z^2.
\end{aligned}$$

$$\begin{aligned}
\gamma(11) &= -(\theta - 1)\mathbb{E}(Z_{t-12}^2) - \theta(1 - \theta)\mathbb{E}(Z_{t-13}^2) \\
&= \sigma_z^2 (1 - \theta)^2.
\end{aligned}$$

$$\begin{aligned}
\gamma(12) &= -\mathbb{E}(Z_{t-12}^2) - (\theta - 1)^2 \mathbb{E}(Z_{t-13}^2) - \theta^2 \mathbb{E}(Z_{t-14}^2) \\
&= \sigma_z^2 [-1 - (1 - 2\theta + \theta^2) - \theta^2] \\
&= -2\sigma_z^2 (1 - \theta + \theta^2).
\end{aligned}$$

$$\begin{aligned}
\gamma(13) &= (1 - \theta)\mathbb{E}(Z_{t-13}^2) + \theta(\theta - 1)\mathbb{E}(Z_{t-14}^2) \\
&= \sigma_z^2 (1 - 2\theta + \theta^2) \\
&= \sigma_z^2 (1 - \theta)^2.
\end{aligned}$$

$$\begin{aligned}
\gamma(14) &= -\theta\mathbb{E}(Z_{t-14}^2) \\
&= \theta\sigma_z^2.
\end{aligned}$$

Using the definition $\rho(h) = \gamma(h)/\gamma(0)$, we see that

$$\rho(h) = \begin{cases} 1 & h = 0; \\ -\frac{(\theta-1)^2}{2(1-\theta+\theta^2)} & |h| = 1; \\ \frac{\theta}{2(1-\theta+\theta^2)} & |h| = 2; \\ \frac{\theta}{4(1-\theta+\theta^2)} & |h| = 10; \\ \frac{(1-\theta)^2}{4(1-\theta+\theta^2)} & |h| = 11; \\ -\frac{1}{2} & |h| = 12; \\ \frac{(1-\theta)^2}{4(1-\theta+\theta^2)} & |h| = 13; \\ \frac{\theta}{4(1-\theta+\theta^2)} & |h| = 14; \\ 0 & \text{otherwise.} \end{cases}$$

Alternative solution: use the autocovariance generating function.

2. Consider a (weakly) stationary $AR(p)$ process, $\{X_t\}$, written

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t.$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 .

(a) Determine the optimal k -step ahead predictor for $k = 1, 2, 3$, for each of the cases $p = 1, 2, 3$.

SOLUTION:

Consider $p = 1$ and $k = 1$. We set

$$(1 - \phi_1 B)\hat{X}_{n+1}(n) = [Z_{n+1}]$$

where

$$[Z_k] = \begin{cases} 0 & \text{if } k > n; \\ Z_k & \text{if } k \leq n. \end{cases}$$

Then

$$\begin{aligned} (1 - \phi_1 B)\hat{X}_{n+1}(n) &= 0 \\ \hat{X}_{n+1}(n) &= \phi_1 \hat{X}_n(n) \\ &= \phi_1 X_n. \end{aligned}$$

Consider $p = 1$ and $k = 2$. We set

$$\begin{aligned}
(1 - \phi_1 B) \hat{X}_{n+2}(n) &= [Z_{n+2}] \\
\implies (1 - \phi_1 B) \hat{X}_{n+2}(n) &= 0 \\
\hat{X}_{n+2}(n) &= \phi_1 \hat{X}_{n+1}(n) \\
&= \phi_1(\phi_1 X_n) \\
&= \phi_1^2 X_n.
\end{aligned}$$

In general, for $p = 1$,

$$\hat{X}_{n+k}(n) = \phi_1^k X_n.$$

Where $p = 2$ and $k = 1$:

$$\begin{aligned}
(1 - \phi_1 B - \phi_2 B^2) \hat{X}_{n+1} &= [Z_{n+1}] \\
\implies \hat{X}_{n+1} &= \phi_1 \hat{X}_n + \phi_2 \hat{X}_{n-1} \\
&= \phi_1 X_n + \phi_2 X_{n-1}
\end{aligned}$$

Where $p = 2$ and $k = 2$:

$$\begin{aligned}
(1 - \phi_1 B - \phi_2 B^2) \hat{X}_{n+2} &= [Z_{n+2}] \\
\implies \hat{X}_{n+2} &= \phi_1 \hat{X}_{n+1} + \phi_2 \hat{X}_n \\
&= \phi_1(\phi_1 X_n + \phi_2 X_{n-1}) + \phi_2 X_n \\
&= (\phi_1^2 + \phi_2) X_n + \phi_1 \phi_2 X_{n-1}
\end{aligned}$$

Where $p = 2$ and $k = 3$:

$$\begin{aligned}
(1 - \phi_1 B - \phi_2 B^2) \hat{X}_{n+3} &= [Z_{n+3}] \\
\implies \hat{X}_{n+3} &= \phi_1 \hat{X}_{n+2} + \phi_2 \hat{X}_{n+1} \\
&= \phi_1 ((\phi_1^2 + \phi_2) X_n + \phi_1 \phi_2 X_{n-1}) + \phi_2 (\phi_1 X_n + \phi_2 X_{n-1}) \\
&= (\phi_1^3 + 2\phi_1 \phi_2) X_n + (\phi_1^2 \phi_2 + \phi_2^2) X_{n-1}
\end{aligned}$$

Where $p = 3$ and $k = 1, 2, 3$, we use the equation

$$\begin{aligned}
(1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3) \hat{X}_{n+k}(n) &= [Z_{n+k}] \\
\implies \hat{X}_{n+k}(n) &= \phi_1 \hat{X}_{n+k-1}(n) + \phi_2 \hat{X}_{n+k-2}(n) + \phi_3 \hat{X}_{n+k-3}(n)
\end{aligned}$$

Hence

$$\hat{X}_{n+1}(n) = \phi_1 X_n + \phi_2 X_{n-1} + \phi_3 X_{n-2}$$

$$\begin{aligned}\hat{X}_{n+2}(n) &= \phi_1(\phi_1 X_n + \phi_2 X_{n-1} + \phi_3 X_{n-2}) + \phi_2 X_n + \phi_3 X_{n-1} \\ &= (\phi_1^2 + \phi_2)X_n + (\phi_1\phi_2 + \phi_3)X_{n-1} + \phi_1\phi_3 X_{n-2}\end{aligned}$$

$$\begin{aligned}\hat{X}_{n+3}(n) &= \phi_1 \hat{X}_{n+2}(n) + \phi_2 \hat{X}_{n+1}(n) + \phi_3 X_n \\ &= \phi_1 ((\phi_1^2 + \phi_2)X_n + (\phi_1\phi_2 + \phi_3)X_{n-1} + \phi_1\phi_3 X_{n-2}) \\ &\quad + \phi_2 (\phi_1 X_n + \phi_2 X_{n-1} + \phi_3 X_{n-2}) + \phi_3 X_n \\ &= (\phi_1^3 + 2\phi_1\phi_2 + \phi_3)X_n + (\phi_1^2\phi_2 + \phi_2^2 + \phi_1\phi_3)X_{n-1} + (\phi_1^2\phi_3 + \phi_2\phi_3)X_{n-2}.\end{aligned}$$

(b) For the $AR(1)$ process, what is the mean squared prediction error?

SOLUTION:

From lectures, we know that the mean squared prediction error for the k -step ahead predictor is

$$\sigma_k^2 = \sigma_z^2 \sum_{j=0}^{k-1} b_j^2$$

where the $\{b_j\}$ terms are coefficients from the $MA(\infty)$ representation of the process. Here

$$\begin{aligned}X_t &= \sum_{j=0}^{\infty} b_j Z_{t-j} \\ &= (1 - \phi_1 B)^{-1} Z_t \\ &= \sum_{j=0}^{\infty} \phi_1^j Z_{t-j}.\end{aligned}$$

Hence

$$\begin{aligned}\sigma_k^2 &= \sigma_z^2 \sum_{j=0}^{k-1} \phi_1^{2j} \\ &= \sigma_z^2 \left(\frac{1 - \phi_1^{2k}}{1 - \phi_1^2} \right) \quad (\text{sum to } k \text{ terms of a GP}).\end{aligned}$$

3. The process $\{X_t\}$ is an $ARIMA(1, 1, 0)$ process, defined

$$(1 - \phi B)\nabla X_t = Z_t$$

where $|\phi| < 1$ and $\{Z_t\}$ is a white noise process with variance σ_z^2 .

Derive the optimal k -step ahead predictor of X_{n+k} , written $\hat{X}_{n+k}(n)$, for this process.

SOLUTION:

First, we derive the predictor for the time series $Y_t = \nabla X_t = X_t - X_{t-1}$ because $\{Y_t\}$ is a weakly stationary $AR(1)$ process. Noting that

$$(1 - \phi B)\hat{Y}_{n+k}(n) = [Z_{n+k}].$$

Where $k = 1$ we obtain

$$\begin{aligned}\hat{Y}_{n+1}(n) - \phi\hat{Y}_n(n) &= 0 \\ \implies \hat{Y}_{n+1}(n) &= \phi Y_n\end{aligned}$$

Similarly,

$$\begin{aligned}\hat{Y}_{n+2}(n) - \phi\hat{Y}_{n+1}(n) &= 0 \\ \implies \hat{Y}_{n+2}(n) &= \phi\hat{Y}_{n+1}(n) \\ &= \phi^2 Y_n.\end{aligned}$$

In general

$$\hat{Y}_{n+k}(n) = \phi^k Y_n.$$

To find $\hat{X}_{n+k}(n)$, consider

$$\begin{aligned}\hat{X}_{n+1}(n) &= X_n + \hat{Y}_{n+1}(n) \\ \hat{X}_{n+2}(n) &= \hat{X}_{n+1}(n) + \hat{Y}_{n+2}(n) \\ &= X_n + \hat{Y}_{n+1}(n) + \hat{Y}_{n+2}(n).\end{aligned}$$

Generally

$$\begin{aligned}\hat{X}_{n+k}(n) &= X_n + \sum_{r=1}^k \hat{Y}_{n+r}(n) \\ &= X_n + \sum_{r=1}^k \phi^r Y_n \\ &= X_n + \left(\frac{\phi(1 - \phi^k)}{1 - \phi} \right) Y_n \\ &= X_n + \left(\frac{\phi(1 - \phi^k)}{1 - \phi} \right) (X_n - X_{n-1}) \\ &= \frac{1 - \phi^{k+1}}{1 - \phi} X_n - \frac{\phi(1 - \phi^k)}{1 - \phi} X_{n-1}.\end{aligned}$$

4. Recall the identity for the optimal k -step ahead predictor

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j} = \sum_{j=0}^{\infty} b_{j+k} Z_{n-j},$$

and define the functions

$$B(\xi) = \sum_{j=0}^{\infty} b_j \xi^j, \quad C_k(\xi) = \sum_{j=0}^{\infty} b_{j+k} \xi^j \quad \text{and} \quad D_k(\xi) = \sum_{j=0}^{\infty} d_j(k) \xi^j.$$

(a) Justify the claim in the lectures that

$$C_k(\xi) = B(\xi) D_k(\xi).$$

SOLUTION:

We have

$$X_t = \sum_{m=0}^{\infty} b_m Z_{t-m} = \sum_{m=0}^{\infty} b_m B^m Z_t.$$

Therefore

$$\begin{aligned} \hat{X}_{n+k}(n) &= \sum_{j=0}^{\infty} d_j(k) X_{n-j} = \sum_{j=0}^{\infty} d_j(k) B^j X_n \\ &= \sum_{j=0}^{\infty} d_j(k) B^j \left(\sum_{m=0}^{\infty} b_m Z_{n-m} \right) \quad \text{because } X_n = \sum_{m=0}^{\infty} b_m Z_{n-m}; \\ &= \sum_{j=0}^{\infty} d_j(k) B^j \sum_{m=0}^{\infty} b_m B^m Z_n \\ &= D_k(\xi) B(\xi) Z_n, \end{aligned}$$

where $\xi = B$, the backward shift operator.

However, the above needs to satisfy

$$\begin{aligned} \hat{X}_{n+k}(n) &= \sum_{j=0}^{\infty} b_{j+k} Z_{n-j} \\ &= \sum_{j=0}^{\infty} b_{j+k} B^j Z_n \\ &= C_k(B) Z_n \end{aligned}$$

which implies that, using notation provided in the lecture notes,

$$D_k(\xi) B(\xi) = C_k(\xi).$$

Hence this identity holds for general ξ and the result is proved.

(b) Why is this identity of importance in prediction?

SOLUTION:

This identity is of importance in prediction because it enables us to express the optimal predictor in terms of a linear combination of present and past values of the *observed* time series $\{X_t\}$, as opposed to a linear combination of past and present *unobserved* white noise terms $\{Z_t\}$.

5. Consider the $AR(3)$ process

$$X_t = 0.3X_{t-1} + 0.2X_{t-2} + 0.1X_{t-3} + Z_t$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 . Suppose that $X_1, \dots, X_n$ have been observed. Determine the predicted values of X_t at $t = n + 1$ and $t = n + 2$.

SOLUTION:

We write

$$\begin{aligned} (1 - 0.3B - 0.2B^2 - 0.1B^3)\hat{X}_{n+k}(n) &= [Z_{n+k}] \\ \implies \hat{X}_{n+k}(n) - 0.3\hat{X}_{n+k-1}(n) - 0.2\hat{X}_{n+k-2}(n) - 0.1\hat{X}_{n+k-3}(n) &= 0. \end{aligned}$$

Then

$$\hat{X}_{n+k}(n) = 0.3\hat{X}_{n+k-1}(n) + 0.2\hat{X}_{n+k-2}(n) + 0.1\hat{X}_{n+k-3}(n)$$

For $k = 1$:

$$\hat{X}_{n+1}(n) = 0.3X_n + 0.2X_{n-1} + 0.1X_{n-2}.$$

For $k = 2$:

$$\begin{aligned} \hat{X}_{n+2}(n) &= 0.3\hat{X}_{n+1}(n) + 0.2X_n + 0.1X_{n-1} \\ &= 0.3(0.3X_n + 0.2X_{n-1} + 0.1X_{n-2}) + 0.2X_n + 0.1X_{n-1} \\ &= 0.29X_n + 0.16X_{n-1} + 0.03X_{n-2}. \end{aligned}$$

6. Suppose that $\{X_t\}$ is an $MA(1)$ process, written

$$X_t = Z_t + \theta Z_{t-1}.$$

The linear predictor of X_{t+1} in terms of $X_t, X_{t-1}, \dots$ is defined

$$\hat{X}_{t+1} = \sum_{r=0}^{\infty} a_r X_{t-r}.$$

Determine the coefficients $\{a_r\}$ such that $\mathbb{E}[(X_{t+1} - \hat{X}_{t+1})^2]$ is minimised. Hence determine the form of $\hat{X}_{t+1}$.

SOLUTION:

Let $\hat{X}_{t+1} = \sum_{r=0}^{\infty} a_r X_{t-r}$ and consider

$$\mathbb{E}[(X_{t+1} - \hat{X}_{t+1})^2].$$

Here

$$\begin{aligned} (X_{t+1} - \hat{X}_{t+1})^2 &= \left(Z_{t+1} + \theta Z_t - \sum_{r=0}^{\infty} a_r (Z_{t-r} + \theta Z_{t-r-1}) \right)^2 \\ &= (Z_{t+1} + (\theta - a_0)Z_t - (a_1 + \theta a_0)Z_{t-1} - (a_2 + \theta a_1)Z_{t-2} - \dots)^2 \\ &= \left(Z_{t+1} + (\theta - a)Z_t - \sum_{r=1}^{\infty} (a_r + \theta a_{r-1})Z_{t-r} \right)^2 \end{aligned}$$

The above term is a square and, hence, when taking the expectation this will be minimised where

$$\begin{aligned} a_0 - \theta &= 0 \\ a_r + \theta a_{r-1} &= 0 \quad \text{for } r = 1, 2, \dots \end{aligned}$$

Then $a_0 = \theta$ and

$$a_r = -\theta a_{r-1}.$$

Hence

$$\begin{aligned} a_1 &= -\theta^2 \\ a_2 &= \theta^3 \\ a_3 &= -\theta^4 \end{aligned}$$

Generally, $a_r = (-1)^r \theta^{r+1}$ and, therefore,

$$\hat{X}_{t+1} = \sum_{r=0}^{\infty} (-1)^r \theta^{r+1} X_{t-r}.$$